

DAVULURI YASHWANT RAO

Full Stack Developer

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PROFESSIONAL SUMMARY

Full Stack Developer with 1.5+ years of experience building production-grade web and mobile applications. Expertise in React, Next.js, Node.js, Python, FastAPI, and cloud deployment. Proven track record optimizing performance, implementing secure authentication/authorization systems, and delivering complete full-stack applications from scratch.

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, Python, C++, SQL

Frontend: Next.js, React.js, React Native, SvelteKit, Svelte, Redux toolkit

Backend: Node.js, Express, FastAPI, Django, Python, Flask

Databases & Storage: PostgreSQL, MongoDB, Firebase, AWS S3, SQLite, DuckDB, MySQL

Infrastructure & DevOps: Docker, Linux, AWS, CI/CD, Git, Postman

APIs & Architecture: REST APIs, GraphQL, JWT Auth, OAuth 2.0, RBAC, Socket.io, Agentic AI, MCP Tools

QA & Testing: Playwright

PROFESSIONAL EXPERIENCE

Full Stack Developer I | Aganitha Cognitive Solutions

Feb 2025 – Present

- **Optimized Python computational pipelines reducing execution time from 20 hours to 4 hours per unit (80% improvement)**

Implemented advanced algorithm optimization and efficient resource allocation for data-intensive processing tasks

- **Built Next.js research data platform from scratch enabling users to save, compare, and validate results against public data sources**

- Architected PostgreSQL backend with Docker containerization and AWS deployment
- Optimized critical performance bottleneck: reduced data retrieval from 40 seconds using request chunking and IndexedDB client-side caching
- Implemented on-demand and 24-hour automated data refresh preventing stale data while minimizing API calls
- Deployed JWT authentication and role-based access control (RBAC) enabling admin data management and user permissions

- **Engineered SvelteKit scientific tool platform over 1.4+ years, evolving from beta-stage to production application**

- Implemented CLI tooling, admin interfaces for formula/algorithm management with PostgreSQL and AWS deployment
- Deployed user and admin role-based access control with JWT authentication
- Continuously delivered feature enhancements, bug fixes, and maintenance updates across full application lifecycle

MERN Full Stack Intern | MyParticipants

Aug 2024 – Nov 2024

- **Developed full-stack features using React, TypeScript, and Node.js with end-to-end ownership from design through production deployment**

- **Implemented JWT and Google OAuth authentication with token-based collaborator access control**

- Enabled secure multi-user collaboration and role-based permissions within MERN application
- **Built comprehensive file management system supporting PDF handling, image processing, report generation, and optimized file uploads**

SELECTED PROJECTS

Xchange – Student Marketplace with Real-Time Communication

React Native | Node.js | Express | Socket.io

- Mobile marketplace enabling peer-to-peer item exchange and commerce within college and hostel communities
- Implemented real-time chat using Socket.io for instant user communication and transaction coordination

VR/AR Real Estate Platform

MERN Stack | Firebase | 3D Visualization | AR/VR

- Real-estate platform with immersive 3D property tours and AR furniture visualization for pre-visit property assessment
- Enabled users to virtually place furniture inside properties to determine fit and spatial arrangement
- [Research published in IEEE paper](#)

Xchange Web – College Community Platform

Next.js | GraphQL | MongoDB

- Purpose-built community platform for college students with community creation, public/private chatrooms, and news sharing
- GraphQL-based data architecture for optimized data fetching and real-time collaboration features

Agentic AI Automation System

Node.js | Next.js | AI Agents | Workplace Automation

- AI agent system automating multi-step workplace workflows including email automation, meeting scheduling, and timesheet management
- Practical Agentic AI implementation focused on executable task automation beyond chatbot interfaces

EDUCATION

B.Tech, Computer Science & Engineering

Vel Tech University | 2021 – 2025 | [CGPA: 8.7](#)

SOCIALS

LinkedIn: www.linkedin.com/in/yashwant-rao-54a506250

GitHub: <https://github.com/Y-A-S-H-W-A-N-T>

Portfolio: <https://yashwant-portfolio.vercel.app/>